

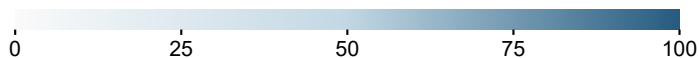
a Fire only

b Matched DeepSets

c Sum GNN

	Terminal						Time average J						Terminal						Time average J						Terminal						Time average J														
	10%			25%			50%			10%			25%			50%			10%			25%			50%			10%			25%			50%			10%			25%			50%		
	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%	10%	25%	50%									
IID (160)	96.9	98.1	99.8	96.7	96.8	99.7				98.8	99.6	99.9	97.7	99.4	99.8				99.0	99.5	99.9	98.1	99.4	99.8				99.0	99.5	99.9	98.1	99.4	99.8				99.0	99.5	99.9	98.1	99.4	99.8			
	98.8	99.9	99.3	96.8	99.1	99.5				99.3	99.9	99.5	98.4	99.6	99.8				99.5	99.9	99.6	98.4	99.7	99.9				99.5	99.9	99.6	98.4	99.7	99.9				99.5	99.9	99.6	98.4	99.7	99.9			
Count OOD (264)																																													
ISO 834 (158)	0.0	0.0	0.0	0.0	0.0	0.0				74.0	77.4	74.8	64.6	82.5	84.3				73.2	75.4	73.9	69.4	79.0	83.5				73.2	75.4	73.9	69.4	79.0	83.5				73.2	75.4	73.9	69.4	79.0	83.5			
ASTM E119 (95)	0.0	0.0	0.0	0.0	0.0	0.0				75.7	78.4	80.6	74.1	82.7	88.0				81.1	81.5	87.1	76.7	82.2	91.6				81.1	81.5	87.1	76.7	82.2	91.6				81.1	81.5	87.1	76.7	82.2	91.6			
Perturbed ISO (111)	34.4	26.4	16.7	6.4	17.0	17.7				74.2	68.8	82.8	75.3	79.9	81.7				77.4	74.2	81.9	82.1	81.6	87.4				77.4	74.2	81.9	82.1	81.6	87.4				77.4	74.2	81.9	82.1	81.6	87.4			
Plateau (103)	98.7	98.0	99.8	98.0	92.1	98.9				98.7	99.3	99.9	97.3	98.9	100.0				99.4	99.6	100.0	98.2	98.8	99.8				99.4	99.6	100.0	98.2	98.8	99.8				99.4	99.6	100.0	98.2	98.8	99.8			
Decay (113)	90.2	96.2	99.7	79.3	87.2	98.5				97.1	98.1	99.7	96.5	95.7	99.0				97.4	98.7	99.6	95.9	96.6	99.6				97.4	98.7	99.6	95.9	96.6	99.6				97.4	98.7	99.6	95.9	96.6	99.6			
Log variant (69)	100.0	98.8	99.6	99.8	98.3	99.3				97.1	100.0	99.7	99.9	100.0	100.0				97.2	100.0	99.8	99.7	100.0	100.0				97.2	100.0	99.8	99.7	100.0	100.0				97.2	100.0	99.8	99.7	100.0	100.0			
External fire (49)	0.0	0.0	0.0	0.0	0.0	0.0				72.0	81.2	83.6	73.5	79.9	93.0				79.2	84.0	89.5	87.6	83.9	88.8				79.2	84.0	89.5	87.6	83.9	88.8				79.2	84.0	89.5	87.6	83.9	88.8			
Bilinear (77)	98.6	94.3	90.2	96.1	96.4	92.1				98.0	98.1	92.5	99.1	96.5	92.3				96.5	96.0	89.5	99.1	97.9	93.5				96.5	96.0	89.5	99.1	97.9	93.5				96.5	96.0	89.5	99.1	97.9	93.5			
Linear (125)	99.5	97.5	98.6	99.3	98.5	99.0				100.0	99.4	99.4	100.0	99.2	99.6				100.0	99.2	99.5	100.0	99.4	99.7				100.0	99.2	99.5	100.0	99.4	99.7				100.0	99.2	99.5	100.0	99.4	99.7			
Smoldering (97)	86.0	100.0	100.0	99.0	100.0	100.0				81.1	100.0	100.0	94.5	100.0	100.0				79.3	100.0	100.0	94.5	100.0	100.0				79.3	100.0	100.0	94.5	100.0	100.0				79.3	100.0	100.0	94.5	100.0	100.0			

Five-seed mean; all 12 protocols × 2 endpoints × 3 case-slot budgets



Available oracle improvement captured (%)

0 = uniform-selection expectation; 100 = retrospective oracle. Gray / NA = undefined ratio.

Expected cutoff-tie selection. Post-review exploratory analysis; no risk or time-saving claim.